Standalone Forecasting Project Report

POS 6933 Tutorial Package

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Executive Summary

This project compares four forecasting models on a shared matched sample across seven forecast horizons.

- **Outcomes:** `log_fatalities_ahead_1m` to `log_fatalities_ahead_7m`.
- **Models:** `regime_rf10`, `regime_stack`, `benchmark_lag123`, and `benchmark_roll3`.
- **Validation:** rolling time-aware cross-validation (5 folds, 12-month validation windows, 120-month minimum initial train window).
- **Main metrics:** MSE, RMSE, deterministic CRPS, probabilistic CRPS, coverage, calibration error, and sharpness.
- **What is newly fit here:** one Random Forest (`regime_rf10`) and one stacked ensemble (`regime_stack`); both are compared against two imported benchmark models.

Modeling Design

- **Regime RF10:** random forest using 10 regime-characteristic predictors.
- **Regime Stack:** stacked ensemble with RF, GBM, and linear model base learners and a linear meta-learner.
- **Benchmark Lag123 and Roll3:** imported benchmark predictions from prior tutorial work, aligned to the same matched evaluation rows.

Performance Tables (One per Model)

Table 1: Benchmark RF (Lag 1-3): performance across horizons 1-7 on matched rows.

H	MSE	RMSE	CRPS Det	CRPS Prob	Cov80	Cov95	CalErr80	Sharp80
1	0.305	0.552	0.218	0.245	0.918	0.954	0.118	1.515
2	0.343	0.586	0.234	0.261	0.917	0.951	0.117	1.596
3	0.362	0.602	0.240	0.269	0.916	0.952	0.116	1.649
4	0.395	0.628	0.255	0.281	0.912	0.945	0.112	1.684
5	0.402	0.634	0.260	0.287	0.909	0.951	0.109	1.730
6	0.414	0.644	0.262	0.289	0.911	0.952	0.111	1.751
7	0.429	0.655	0.268	0.292	0.911	0.950	0.111	1.748

Table 2: Benchmark RF (Rolling 3-Month Avg): performance across horizons 1-7 on matched rows.

H	MSE	RMSE	CRPS Det	CRPS Prob	Cov80	Cov95	CalErr80	Sharp80
1	0.396	0.629	0.250	0.292	0.899	0.941	0.099	1.726
2	0.456	0.675	0.268	0.309	0.898	0.942	0.098	1.796
3	0.470	0.686	0.273	0.315	0.896	0.942	0.096	1.842
4	0.504	0.710	0.284	0.325	0.895	0.941	0.095	1.892
5	0.525	0.725	0.290	0.331	0.896	0.941	0.096	1.915
6	0.532	0.729	0.291	0.333	0.897	0.941	0.097	1.938
7	0.536	0.732	0.295	0.334	0.896	0.942	0.096	1.937

Table 3: Regime Ensemble (RF+GBM+LM): performance across horizons 1-7 on matched rows.

H	MSE	RMSE	CRPS Det	CRPS Prob	Cov80	Cov95	CalErr80	Sharp80
1	0.662	0.813	0.363	0.377	0.888	0.938	0.088	2.075
2	0.765	0.875	0.385	0.406	0.878	0.933	0.078	2.183
3	0.886	0.941	0.419	0.433	0.878	0.935	0.078	2.286
4	0.960	0.980	0.423	0.443	0.873	0.934	0.073	2.289
5	1.014	1.007	0.437	0.460	0.872	0.929	0.072	2.339
6	1.025	1.012	0.422	0.461	0.876	0.936	0.076	2.427
7	1.035	1.017	0.429	0.465	0.877	0.934	0.077	2.443

Table 4: Regime Random Forest (10 vars): performance across horizons 1-7 on matched rows.

H	MSE	RMSE	CRPS Det	CRPS Prob	Cov80	Cov95	CalErr80	Sharp80
1	0.695	0.834	0.409	0.393	0.881	0.938	0.081	2.106
2	0.810	0.900	0.447	0.426	0.872	0.933	0.072	2.224
3	0.939	0.969	0.479	0.454	0.870	0.933	0.070	2.334
4	1.003	1.001	0.489	0.465	0.869	0.935	0.069	2.368
5	1.087	1.042	0.511	0.485	0.870	0.929	0.070	2.439
6	1.114	1.055	0.523	0.499	0.876	0.938	0.076	2.616
7	1.130	1.063	0.518	0.501	0.880	0.937	0.080	2.637

How to read these tables

- Lower values are better for MSE, RMSE, CRPS, calibration error, and sharpness.
- Coverage columns should be close to 0.80 and 0.95 for well-calibrated uncertainty.
- H is the forecast horizon (months ahead).

Predictor Variables

Table 5: Regime-characteristic predictor variables used in the modeling pipeline.

Variable	Source	What it measures	Typical range
vdem_v2x_polyarchy	V-Dem	Electoral democracy quality.	0 to 1
vdem_v2x_libdem	V-Dem	Liberal democracy and institutional constraints.	0 to 1
vdem_v2x_partipdem	V-Dem	Participatory depth of democracy.	0 to 1
vdem_v2x_delibdem	V-Dem	Deliberative quality of decision-making.	0 to 1
vdem_v2x_egaldem	V-Dem	Egalitarian distribution of political power.	0 to 1
wgi_VA_EST	WGI	Voice and accountability.	about -2.5 to 2.5
wgi_GE_EST	WGI	Government effectiveness.	about -2.5 to 2.5
wgi_RL_EST	WGI	Rule of law quality.	about -2.5 to 2.5
fh_cl	Freedom House	Civil liberties rating.	1 to 7
polity_polity2	Polity5	Regime authority spectrum score.	-10 to 10

Figures

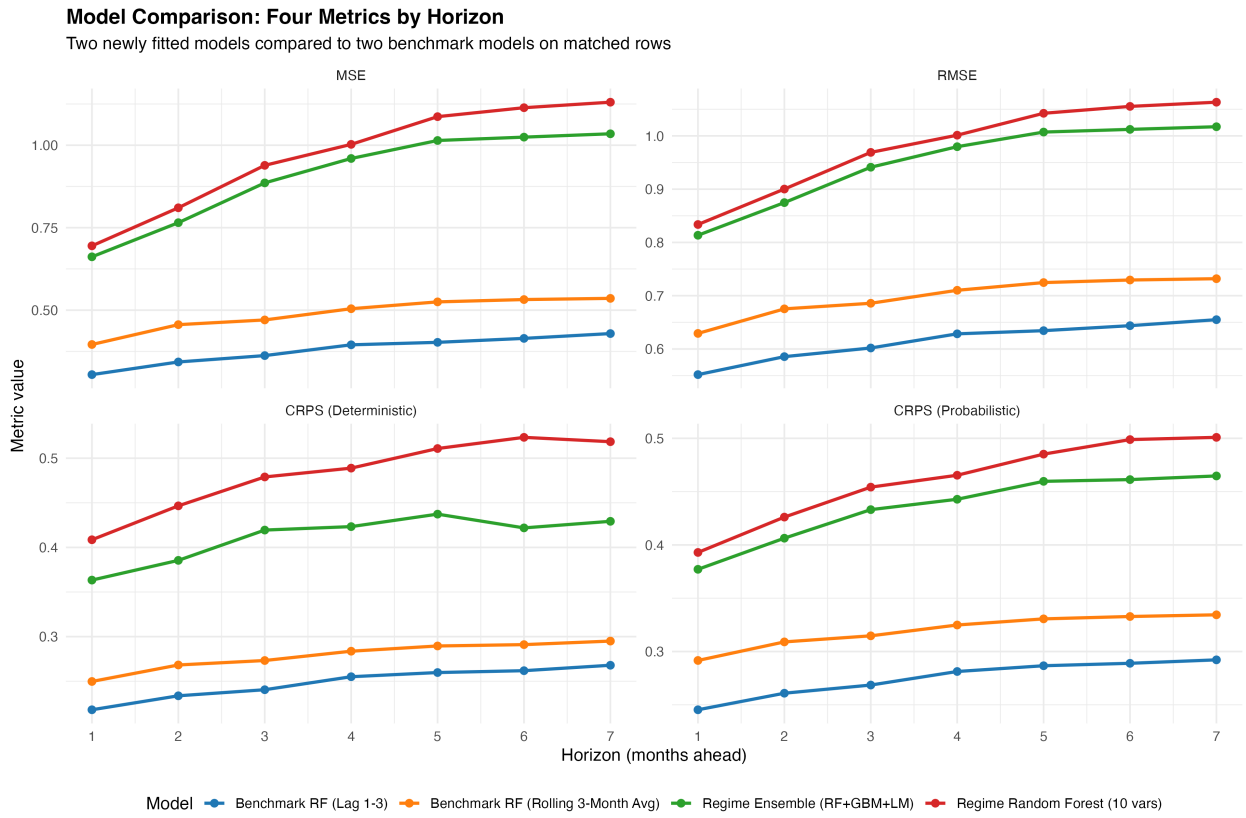

Figure 1: Core comparison metrics by horizon across all four models.

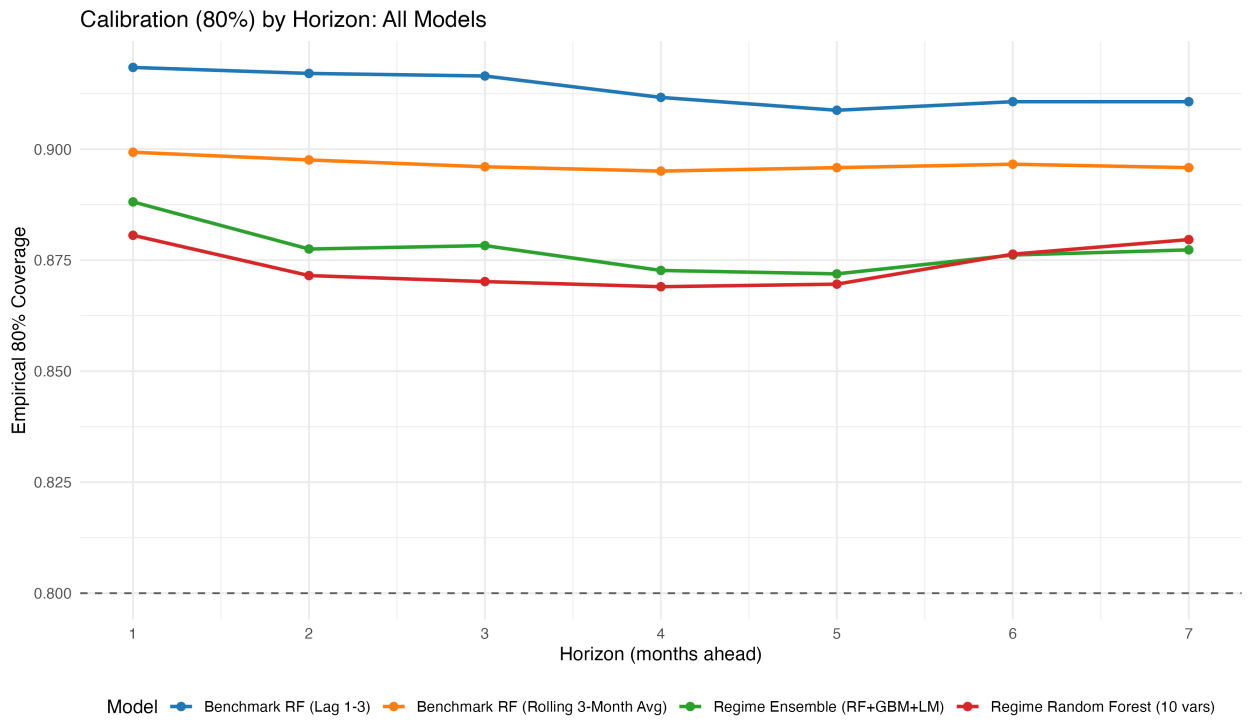

Figure 2: Empirical 80% coverage by horizon across models (dashed line = nominal target).

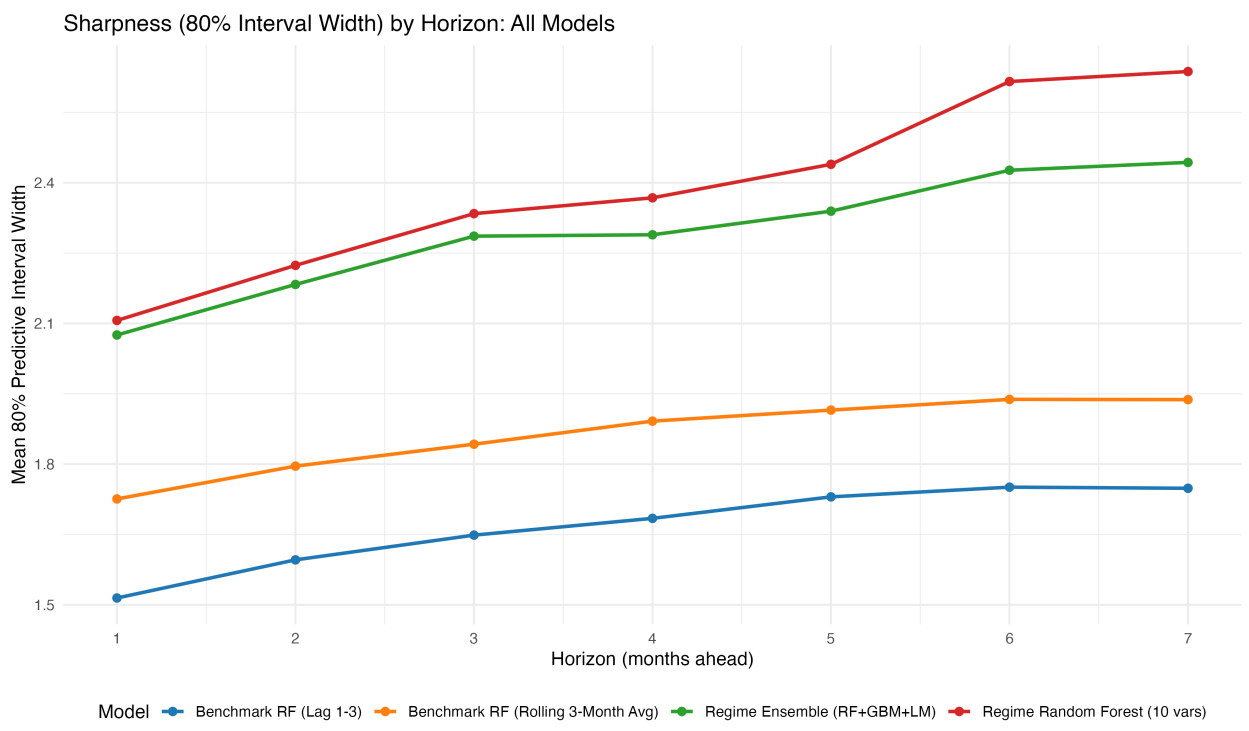

Figure 3: 80% predictive interval width (sharpness) by horizon across models.

Figure 4: Predicted vs actual values on matched test rows, faceted by model.

Figure 5: Mean prediction paths by horizon; black line is the observed mean path.

Reproducibility

- Main script: `code/forecasting_project_all_in_one.R`
- This report reads auto-generated LaTeX tables from: `results/tables`